Size::148x210

----------IDAMAN PHARMA MANUFACTURING ---------

SYNAMOX FOR ORAL SUSPENSION 125MG/5ML

COMPOSITION

Each 5 ml reconstituted suspension contains Amoxicillin Trihydrate equivalent to 125 mg Amoxicillin.

DESCRIPTION

Dry Powder:
An off-white to slightly yellow, fine, free flowing, homogeneous powder, free from lumps and substantially free from dark specks.
Colour: Off-white to slightly yellow.
Odour: Characteristic of penicillin.

Reconstitution Oral Suspension:
An off-white to slightly yellow, free flowing homogeneous suspension.
Colour: Off-white to slightly yellow.
Odour: Fruity.
Taste: Sweet.

PHARMACODYNAMICS

Amoxicillin is bactericidal; its action depends on its ability to reach and bind penicillin-binding proteins (PBP-1 and PBP-3) located in bacterial cytoplasmic membranes; Amoxicillin inhibits bacterial septum and cell wall synthesis, probably by acylation of membrane-bound transpeptidase enzymes. This prevents cross-linkage of peptidoglycan chains which is necessary for bacterial cell wall strength and rigidity. Also, cell division and growth are inhibited and lysis and elongation of susceptible bacteria frequently occur. Rapidly dividing bacteria are those most susceptible to the action of amoxicillin.

PHARMACOKINETICS

Absorption Amoxicillin is stable in the presence of gastric acid and rapidly absorbed from the gut to an extent of 72 to 93%. Absorption is independent of food intake. Peak blood levels are achieved 1 to 2 hours after administration. After 250 and 500mg doses of amoxicillin, average peak serum concentrations of 5.2 mcg/ml and 8.3 mcg/ml respectively have been reported.

Distribution Amoxicillin is not highly protein bound. Approximately 18% of total plasma drug content is bound to protein. Amoxicillin diffuses readily into most body tissues and fluids, including sputum and saliva but not the brain and spinal fluid. Inflammation generally increases the permeability of the meninges to penicillins and this may apply to amoxicillin. Amoxicillin diffuses across the placenta and a small percentage is excreted into the breast milk.

Metabolism Amoxicillin is excreted mainly via the urine where it exists in a high concentration. Amoxicillin is also partly excreted in the urine as the inactive penicilloic acid in quanttes equivalent to 10 to 25% of the initial dose. Small amounts of the drug are also excreted in faeces and bile. Concentrations in the bile may vary and are dependent upon normal biliary function.

Excretion Approximately 60 to 70% of amoxicillin is excreted unchanged in urine during the first 6 hours after administration of a standard dose. The elimination half-life is approximately 1 hour. Concurrent administration of probenecid delays amoxicillin excretion. In patients with end stage renal failure, the half-life ranges between 5 to 20 hours. The substance is haemodialysable.

INDICATIONS

Amoxicillin is indicated in the treatment of:

i. Ear, nose and throat infections caused by Streptococci, Pneumococci, non penicillinase-producing Staphylococci and *Haemophilus influenza*.
ii. Genitourinary tract infections caused by *Escherichia coli*, *Proteus mirabilis* and *Streptococcus faecalis*.
iii. Acute uncomplicated anogenital and urethral gonorrhaea in males and females caused by strains of *Neisseria gonorrhoea* sensitive to amoxicillin.
iv. Skin and soft tissues infections caused by Streptococci, non penicillinase- producing Staphylococci, *Escherichia coli* and *Proteus mirabilis*.

DOSAGE AND ADMINISTRATION

Usual Paediatric Dose (Oral):
Infants up to 6kg of body weight: The equivalent of anhydrous Amoxicillin: 25mg to 50mg every 8 hours. Infants 6 to 8 kg of body weight: 50mg to 100mg every 8 hours. Infants and children 8 to 20kg of body weight: 6.7mg to 13.3mg per kg of body weight every 8 hours. Children 20kg of body weight and over: May follow usual adult doses.

Note: Even though some infants and children may require a larger dose than adults on a weight basis, the total daily adult dose should not be exceeded in infants and children.

Gonorrhoea:
Oral, equivalent of anhydrous Amoxicillin - 50mg per kg of body weight and 25mg of Probenecid per kg of body weight simultaneously as a single dose in prepubertal children. Probenecid is not recommended in children under 2 years of age.

ROUTE OF ADMINISTRATION

Oral

INSTRUCTION FOR USE

RECONSTITUTION FOR SUSPENSION

For 60 mL: Shake bottle to loosen the powder. Add distilled water or boiled water (cooled to room temperature) slightly below the 60 ml mark and shake vigorously. Make up the volume to the 60 ml mark. Shake well.

i. Take this medicine at regular intervals (on hourly basis) and the prescribed course should be completed even when you feel better.
ii. Do not miss the doses prescribed.
iii. This medicine may be taken on full or empty stomach.
iv. Keep this medicine out of reach of children.
v. Check with your doctor if no improvement within a few days.

CONTRAINDICATIONS

Hypersensitivity to the active substance, to any of the penicillins or to any of the excipients. History of a severe immediate hypersensitivity reaction (e.g. anaphylaxis) to another beta-lactam agent (e.g. a cephalosporin, carbapenem or monobactam).

ADVERSE EFFECTS

Infections and infestations:
Very rare: Mucocutaneous candidiasis.

Blood and lymphatic system disorders:
Very rare: Reversible leucopenia (including severe neutropenia and agranulocytosis), reversible thrombocytopenia, haemolytic anaemia and prolongation of bleeding time and prothrombin time (See Warnings and Precautions section).

Immune system disorders:
Very rare: Severe allergic reactions including angioneurotic oedema, anaphylaxis, serum sickness and hypersensitivity vasculitis.
Not Known: Jarisch-Herxheimer reaction.

Nervous system disorders:
Very rare: Hyperkinesia, dizziness and convulsions.

Gastrointestinal disorders:
Common: Diarrhoea and nausea.
Uncommon: Vomiting.
Very rare: Antibiotic-associated colitis (including pseudomembranous colitis and haemorrhagic colitis. black hairy tongue and superficial tooth discolouration. Superficial tooth discolouration has been reported in children.

Hepatobiliary disorders:
Very rare: Hepatitis and cholestatic jaundice. A moderate rise in AST and/or ALT.

Skin and subcutaneous tissue disorders:
Common: Skin rash.
Uncommon: Urticaria and pruritus.
Very rare: Skin reactions such as erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis, bullous and exfoliative dermatitis, acute generalised exanthematous pustulosis (AGEP), Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS).

Renal and urinary tract disorders:
Very rare: Interstitial nephritis and crystalluria.

WARNINGS AND PRECAUTION

Hypersensitivity reactions
Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Synamox for Oral Suspension, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, Synamox for Oral Suspension must be discontinued immediately and appropriate alternative therapy instituted.

Non-susceptible microorganisms
Amoxicillin is not suitable for the treatment of some types of infection unless the pathogen is already documented and known to be susceptible or there is a very high likelihood that the pathogen would be suitable for treatment with amoxicillin. This particularly applies when considering the treatment of patients with urinary tract infections and severe infections of the ear, nose and throat.

Prescribing amoxicillin in the absence of a proven or strongly suspected bacterial infection or a prophylactic indication is unlikely to provide benefit to the patient and increases the risk of the development of drug-resistant bacteria.

Convulsions
Convulsions may occur in patients with impaired renal function or in those receiving high doses or in patients with predisposing factors (e.g. history of seizures, treated epilepsy or meningeal disorders).

Renal impairment
In patients with renal impairment, the dose should be adjusted accordingly to the degree of impairment

Skin reactions
The occurrence at the treatment initiation of a feverish generalised erythema associated with pustula may be a symptom of acute generalised exanthemous pustulosis (AGEP). This reaction requires amoxicillin discontinuation and contraindicates any subsequent administration. Amoxicillin should be avoided if infectious mononucleosis is suspected since the occurrence of a morbilliform rash has been associated with this condition following the use of amoxicillin.

Overgrowth of non-susceptible microorganisms
Prolonged use may also occasionally result in overgrowth of non-susceptible organisms. Antibiotic-associated colitis has been reported with nearly all antibacterial agents and may range in severity from mild to life threatening. Therefore, it is important to consider this diagnosis in patients who present with diarrhoea during or subsequent to the administration of any antibiotics. Should antibiotic-associated colitis occur, amoxicillin should immediately be discontinued, a physician consulted and an appropriate therapy initiated. Appropriate fluid and electrolyte management, protein supplementation, antibiotic treatment of *C. difficile*, and surgical evaluation should be instituted as clinically indicated. Anti-peristaltic medicinal products are contraindicated in this situation.

Prolonged therapy
Periodic assessment of organ system functions; including renal, hepatic and haematopoietic function is advisable during prolonged therapy. Elevated liver enzymes and changes in blood counts have been reported.

Jarisch-Herxheimer reaction
The Jarisch-Herxheimer reaction has been seen following amoxicillin treatment of Lyme disease. It results directly from the bactericidal activity of amoxicillin on the causative bacteria of Lyme disease, the spirochaete *Borrelia burgdorferi*. Patients should be reassured that this is a common and usually self-limiting consequence of antibiotic treatment of Lyme disease.

Crystalluria
In patients with reduced urine output, crystalluria has been observed very rarely, predominantly with parenteral therapy. During the administration of high doses of amoxicillin, it is advisable to maintain adequate fluid intake and urinary output in order to reduce the possibility of amoxicillin crystalluria. In patients with bladder catheters, a regular check of patency should be maintained.

Interference with diagnostic tests
Elevated serum and urinary levels of amoxicillin are likely to affect certain laboratory tests. Due to the high urinary concentrations of amoxicillin, false positive readings are common with chemical methods. It is recommended that when testing for the presence of glucose in urine during amoxicillin treatment, enzymatic glucose oxidase methods should be used. The presence of amoxicillin may distort assay results for oestriol in pregnant women.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

No studies on the effects on the ability to drive and use machines have been performed. However, undesirable effects may occur (e.g. allergic reaction, dizziness, convulsions), which may influence the ability to drive and use machine.

INTERACTION WITH OTHER MEDICAMENTS

Probenecid: Concomitant use of probenecid is not recommended. Probenecid decreases the renal tubular secretion of amoxicillin. Concomitant use of probenecid may result in increased and prolonged levels of amoxicillin.

Allopurinol: Concurrent administration of allopurinol during treatment with amoxicillin can increase the likelihood of allergic skin reactions.

Chloramphenicol, macrolides, sulfonamides and tetracyclines: Since bacteriostatic drugs may interfere with the bactericidal effect of amoxicillin in the treatment of situations where rapid bactericidal effect is necessary, it is best to avoid concurrent therapy.

Methotrexate: Penicillins may reduce the excretion of methotrexate causing a potential increase in toxicity.

Oral typhoid vaccine: The oral typhoid vaccine is inactivated by antibacterials.

Oral anticoagulants: Oral anticoagulants and penicillin antibiotics have been widely used in practice without reports of interaction. However, in the literature there are cases of increased international normalised ratio in patients maintained on acenocoumarol or warfarin and prescribed a course of amoxicillin. If co-administration is necessary, the or ratio should be carefully monitored with the addition or withdrawal of amoxicillin. Moreover, adjustments in the dose of oral anticoagulants may be necessary.

Oral estrogen/progesterone contraceptives: In common with other antibiotics, amoxicillin may affect the gut flora, leading to lower estrogen reabsorption and reduced efficacy of combined oral estrogen/progesterone contraceptives.

PREGNANCY AND LACTATION

Pregnancy: Limited data on the use of amoxicillin during pregnancy in humans do not indicate an increased risk of congenital malformations. Amoxicillin may be used in pregnancy when the potential benefits outweigh the potential risks associated with treatment.

Breast-feeding: Amoxicillin is excreted into breast milk in small quantities with the possible risk of sensitisation. Consequently, diarrhoea and fungus infection of the mucous membranes are possible in the breast-fed infant, so that breast-feeding might have to be discontinued. Amoxicillin should only be used during breastfeeding after benefit/risk assessment by the physician in charge.

OVERDOSE AND TREATMENT

Symptoms and signs of overdose: Gastrointestinal symptoms such as nausea, vomiting and diarrhoea and disturbance of the fluid and electrolyte balances may be evident. Amoxicillin crystalluria, in some cases leading to renal failure has been observed. Convulsions may occur in patients with impaired renal function or in those receiving high doses.

Treatment of intoxication: Gastrointestinal symptoms may be treated symptomatically, with attention to the water/electrolyte balance. Amoxicillin may be removed from the circulation by haemodialysis.

STORAGE CONDITIONS

Keep container tightly closed in a dry place, store below 30°C prior to reconstitution and refrigerate after reconstitution.
Protect from light and freezing.

SHELF LIFE

2 years from the date of manufacture.

PRESENTATION

(Malaysia Reg. No.: MAL 19890250AZ)
Packed in plastic containers of 60 ml.

For further information, please consult your doctor or your pharmacist.

Date of Revision: 18 Dec 2020

PRODUCT REGISTRATION HOLDER/ MANUFACTURER:
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